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Optimal nitrogen management for high yield and N use efficiency of ratoon sorghum

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Sorghum ratooning, a time and labor-saving cultivation practice, is increasingly being adopted by farmers in Southwest China as an alternative. Efficient N fertilizer management is critical for economical production of sorghum and the long-term protection of the environment. To investigate the impact of N management on grain yield and nitrogen use efficiencies (NUEs) of ratoon sorghum system, a three-year field experiment was conducted for Jinyunuo3 (a hybrid cultivar) and Guojiaohong1 (an inbred cultivar) using 12 combinations of N rates and splitting ratios. The results showed that increasing N rate and splitting application times led to improvements in various growth parameters such as dry matter weight, crop growth rate (CGR), leaf area index (LAI), and photosynthetic potential (PP). The main, ratoon, and annual yields increased with N rate increase, but there was no significant difference between 225 and 150 kg N ha^{-1} in the ratoon and annual yields. Splitting the application of N fertilizer enhanced grain yield compared to a single dose application method, especially three-split applications yielded higher than two-split applications. Compared with N rates of 225 and 150 kg ha^{-1}, N rate of 75 kg ha^{-1} increased apparent recovery rate of applied nitrogen (RE_N), agronomic efficiency of applied nitrogen (AE_N), and partial factor productivity from applied nitrogen (PFP_N) in both main season and whole year. But through splitting application methods at high N rates could achieve similar or even higher levels of NUEs compared to all applied as basal fertilizer at low N rates. Therefore, it could be recommended that applying 150 kg N ha^{-1} with a basal-jointing-heading fertilizer ratio of 2:4:4 represented an efficient N management practice to synchronously obtain high grain yield and NUEs in ratoon sorghum system in Southwest China.

Keywords Sorghum ratooning, Nitrogen management, Grain yield, Nitrogen use efficiency

Sorghum (*Sorghum bicolor*), the world's fifth most important cereal, is primarily utilized for liquor production in China[1]. Southwest China is an advantageous region for high-quality brewing sorghum (waxy), and the yield of sorghum in this area is crucial for the liquor-making industry[2]. In recent years, ratoon sorghum, a practice of harvesting grain from tillers originating from the stubble of a previously harvested crop (main crop, MC)[3], has attracted farmers' attention in this region[4]. The ratoon crop (RC) has a shorter growth period of at least 20 days compared to directly sown sorghum[5], thereby increasing land productivity in Southwest China where thermal energy exceeds that required by single-season sorghum but falls short for double-season sorghum[6]. Land preparation and crop establishment (direct-seeding or transplanting) are unnecessary for RC, significantly reducing labor input[7].

Nitrogen (N) application is one of the most important managements to obtain optimal yield of RC, and there are distinct differences between the fertilization management for ratoon sorghum system and single-cropping sorghum[8]. However, there is limited knowledge regarding the impact of N on ratoon sorghum system, with insufficient information available about the rate, frequency, and timing of fertilizer application[9]. In China, the N application method for ratoon sorghum was copied the operation of rice, that is, in addition to the normal fertilization in the main season, applying two N fertilizers for promoting bud development (N_{bud}) and promoting the development of regenerated tiller (N_{tiller})[10,11]. N_{bud} is applied around the midpoint between heading and harvest of MC to increase the green leaf area and prevent death of ratooning buds after full heading stage[12]. N_{tiller} is applied immediately after harvest of MC[07], to improve the effective tillering percentage and grain number per ear[13]. However, the sorghum cultivar chosen for ratoon practice is usually stay-green hybrid, exhibiting greener

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leaves and stems during the grain-filling[14]. Stay-green trait, combined with a strong root system of sorghum, could support the development of regenerate buds for second, third, or more growth of ratoon crops; moreover, only one tiller from stubble was left to grow to mature for RC[9], but in our previous study[15], the number of regenerate new tillers could reach to 7.5 and the length was up to 40 cm 10 d after MC cut. According to the above mentioned factors, we considered that it was unnecessary to apply N_{bud} or N_{tiller} to improve bud growth and yield of ratoon sorghum.

Ardiyanti et al.[9] suggested that the fertilizer dosage for the growth of RC should be equivalent to that of the MC, applied three weeks after cutting the stem. Similarly, Escalada and Plucknett[16] adopted a similar approach by applying equal amounts of N on both MC and RC. However, it is worth noting that the yield of ratoon sorghum is often lower compared to that of main sorghum[17], which implies that applying an identical amount of N fertilizer would result in lower N use efficiencies (NUEs). Mourtzinis et al.[18], in their study, conducted experiments with starter fertilizer treatment at planting and additional N treatment immediately following the first grain harvest; however, they found no significant effect of starter fertilizer on grain yield. Faesal et al.[19], on the other hand, fertilized MC with 135 kg N ha^{-1} and applied after the regenerate tillers begin to grow with a rate of 50% of the MC fertilizing dose.

It appears that there is variable information in the literature about the fertilization for RC, and there is currently no available information specific for Southwest China. In order to establish an optimal N management strategy for RC in this region, it becomes imperative for us first determine how N application in MC affects yield and N use efficiency of ratooning sorghum system. Our previous study demonstrated that the recommended N rate ranging between 150 and 225 kg ha^{-1}[20,21] exceeded requirement for main sorghum production, suggesting a postponed N application in main season to improve ratoon sorghum yield[22]. Therefore, we conducted field experiments with various N management treatments for MC and measured the yield and NUEs of ratooning sorghum system. We believe that the knowledge obtained through this study would serve as a guide for optimizing nitrogen application to achieve a balance of high yield and high efficiency of ratoon sorghum system.

Results

Growth parameters

Crop growth duration

The life-cycle duration and phenological events were not significantly different between the two cultivars, therefore the N-splitting treatment and sampling were conducted on the same day for the two cultivars (Table 1). Total growth durations of RC were shorter than those of MC by 9.45%, 15.32%, and 8.13% in 2019, 2020, and 2021, respectively. The variation on total growth duration of RC among 3 years was greater than that of MC, mainly due to the biggest variation on heading to maturity stage of RC (HSRC-MSRC).

Dry matter accumulation

The dry matter weight of each stage was significantly affected by cultivar (C), treatment (T), and their interaction (Table 2). The dry matter reached its maximum at maturity stage of MC (MSMC). The dry matter weight of each stage of MC was higher than the corresponding stage of RC (Fig. 1). Nitrogen application improved dry

Year	S-JSMC	JSMC-HSMC	HSMC-MSMC	MC	MSMC-HSRC	HSRC-MSRC	RC	Whole season
2019	76	18	33	127	65	50	115	242
2020	69	20	35	124	70	35	105	229
2021	68	20	35	123	68	45	113	236

Table 1. Growth durations (d) in the main, ratoon, and the whole seasons of sorghum in 2019, 2020, and 2021. S: sowing; JSMC, jointing stage of MC; HSMC, heading stage of MC; MSMC, maturity stage of MC; HSRC, heading stage of RC; MSRC, maturity stage of RC.

Variation source	Total dry matter weight				
	JSMC	HSMC	MSMC	HSRC	MSRC
Y	0.09	16.15**	40.43**	13.18**	46.27**
C	62.44**	4426.25**	3635.75**	5479.98**	3348.33**
T	42.80**	168.28**	204.55**	184.40**	199.29**
Y×C	0.15	1.01	15.12**	7.80**	11.61**
Y×T	1.57	0.96	1.13	2.79**	2.24**
C×T	2.69**	14.42**	9.35**	28.73**	31.65**
Y×C×T	1.61	1.02	0.67	0.88	1.04

Table 2. Variance analysis (ANOVA) of total dry matter weight (g plant^{-1}) on different growth stages for the year (Y), cultivar (C), N management (T). JSMC, jointing stage of MC; HSMC, heading stage of MC; MSMC, maturity stage of MC; HSRC, heading stage of RC; MSRC, maturity stage of RC. *$P<0.05$; **$P<0.01$.

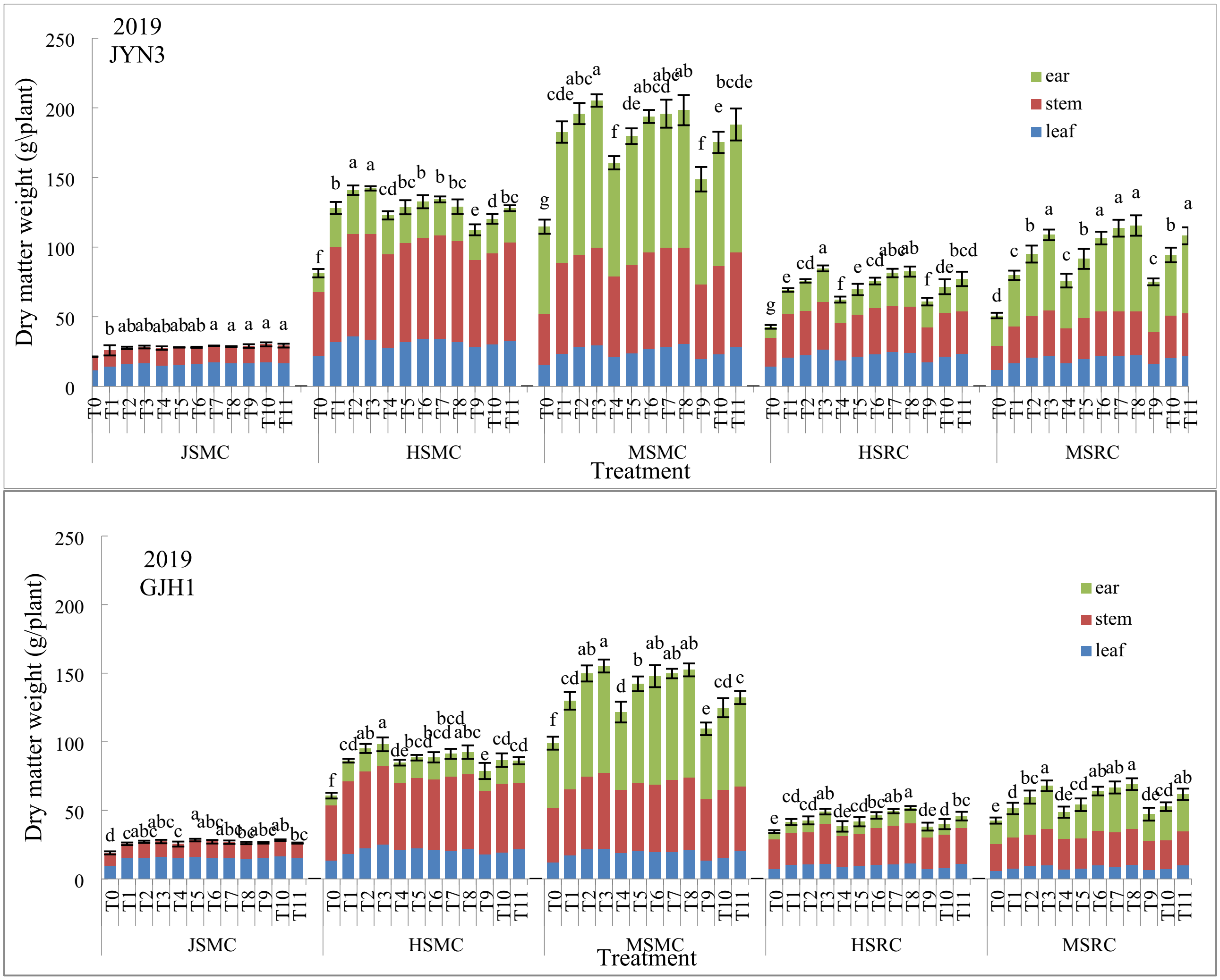

Fig. 1. The changes of dry matter accumulation at different growth stages of MC and RC. Error bars represent the standard error of the sample mean (n = 3). Different letters indicate differences ($P<0.05$). JSMC, jointing stage of MC; HSMC, heading stage of MC; MSMC, maturity stage of MC; HSRC, heading stage of RC; MSRC, maturity stage of RC.

matter accumulation for each growth stage. Except at jointing stage, dry matter weight increased with the N rate increasing at all growth stages. N splitting enhanced dry matter accumulation, and the highest dry matter weights for MSMC and maturity stage of RC (MSRC) were both observed in T3 condition.

Crop growth rate

Crop growth rate (CGR) of each growth period was significantly affected by Year (Y), cultivar (C), treatment (T), and C × T interaction (Table 3). Nitrogen application improved CGR for each growth period, and CGRs of jointing to heading stage of main crop (JSMC-HSMC), heading to mature stage of main crop (HSMC-MSMC), and mature stage of main crop to heading stage of ratoon crop (MSMC-HSRC) increased with N rate increasing. But for heading to mature stage of ratoon crop (HSRC-MSRC), the highest CGR was observed under N rate of 150 kg ha^{-1}. Under a given N rate 225, 150, and 75 kg ha^{-1}, splitting N application remarkably promoted CGR for all periods except sowing to jointing stage of main crop (S-JSMC), and three-split pattern resulted in a higher CGR than two-split pattern.

Leaf area index

Leaf area index (LAI) for each growth stage was significantly affected by Year (Y), cultivar (C), and treatment (T), and their interactions had significant effects on LAI for all growth stages except JSMC (Table 4). The LAI reached the highest at HSMC, and LAI of MC was higher than that of RC. High N rate increased LAI significantly, especially enhanced LAI for RC. All N applied as basal fertilizer produced the highest LAI for JSMR, while splitting N application resulted in a higher LAI on all the other stages. Increasing the ratio of heading-N fertilizer significantly increased the LAI for MSMC, HSRC, and MSRC at each N rate.

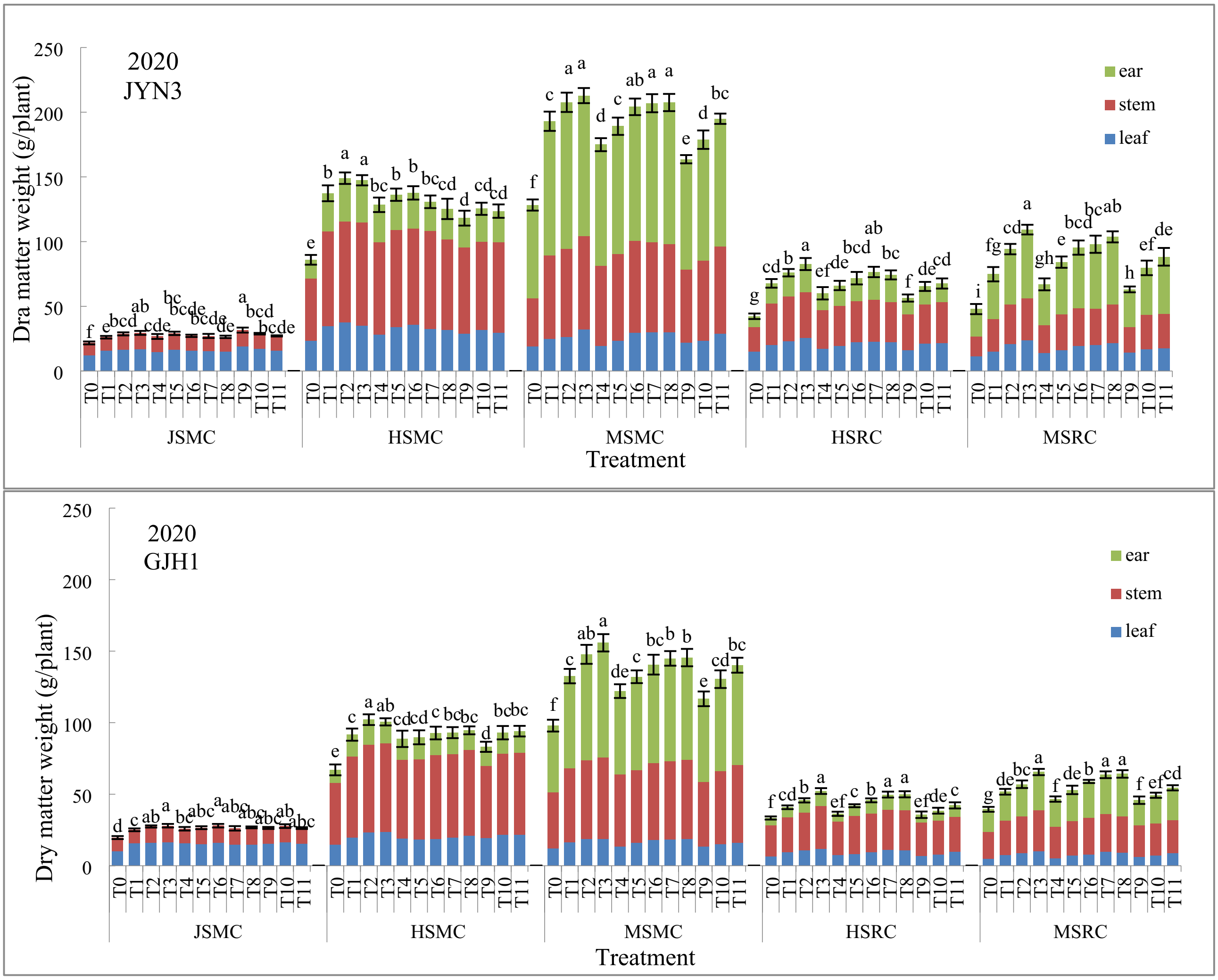

Fig. 1. (continued)

Photosynthetic potential
There were significant effects of cultivar (C), treatment (T), and their interactions on photosynthetic potential (PP) for each growth period (Table 5). N application improved PP for each growth period and had a greater effect on PP of RC than MC. Under a given N rate of 225, 150, and 75 kg ha^{-1}, splitting N application remarkably promoted PP for HSMC-MSMC, MSMC-HSRC, and HSRC-MSRC, and three-split pattern resulted in a higher PP than two-split pattern.

Main, ratoon and annual yields

The main, ratoon and annual yields were significantly affected by year (Y), cultivar (C), treatment (T), and their interactions (Table 6). Average across 3 years, the main, ratoon, and annual yields were 5599.00, 3135.90 and 8735.00 kg ha^{-1} for JYN3, and 4134.95, 1047.62, and 5182.57 kg ha^{-1} for GJH1, respectively.

Nitrogen fertilizer enhanced grain yield of both cultivars in all 3 years, especially enhanced ratoon yield. N rates of 225, 150, and 75 kg ha^{-1} improved main yield by 49.31–54.24%, 41.11–47.84%, and 32.65–35.69% compared with T0 for JYN3, and 66.41–72.67%, 52.97–63.88%, and 39.81–42.80% for GJH1. As for ratoon yield, N rates of 225, 150, and 75 kg ha^{-1} increased 84.75–110.27%, 85.11–104.31%, and 64.60–69.90% compared with T0 for JYN3, and 77.76–94.75%, 54.34–93.08, and 33.80–54.67% for GJH1. The annual yields of N rates of 225, 150, and 75 kg ha^{-1} were 61.07–69.86%, 58.93–63.93% and 43.15–45.28% more than that of T0 for JYN3, and 71.73–73.71%, 59.90–64.82% and 41.11–43.46% for GJH1. The main, ratoon, and annual yields under 225 and 150 kg N ha^{-1} were significantly higher than those under 75 kg N ha^{-1}. Except for the main yields of JYN3 in 2020 and GJH1 in 2021, there was no significant difference between the 225 and 150 kg N ha^{-1} in the main, ratoon, and annual yields.

The N application method had a significant effect on grain yield, and splitting application increased the main, ratoon, and annual yields under a given N rate. The increments of the ratoon yield caused by splitting application were greater than those of the main yield. Three-split application under a given N rate of 225, 150, and 75 kg ha^{-1} resulted in a higher main, ratoon, and annual yield than two-split application. The main, ratoon, and annual yields under T6 were not significantly different from those under T1, and even higher than those under T1. Except the

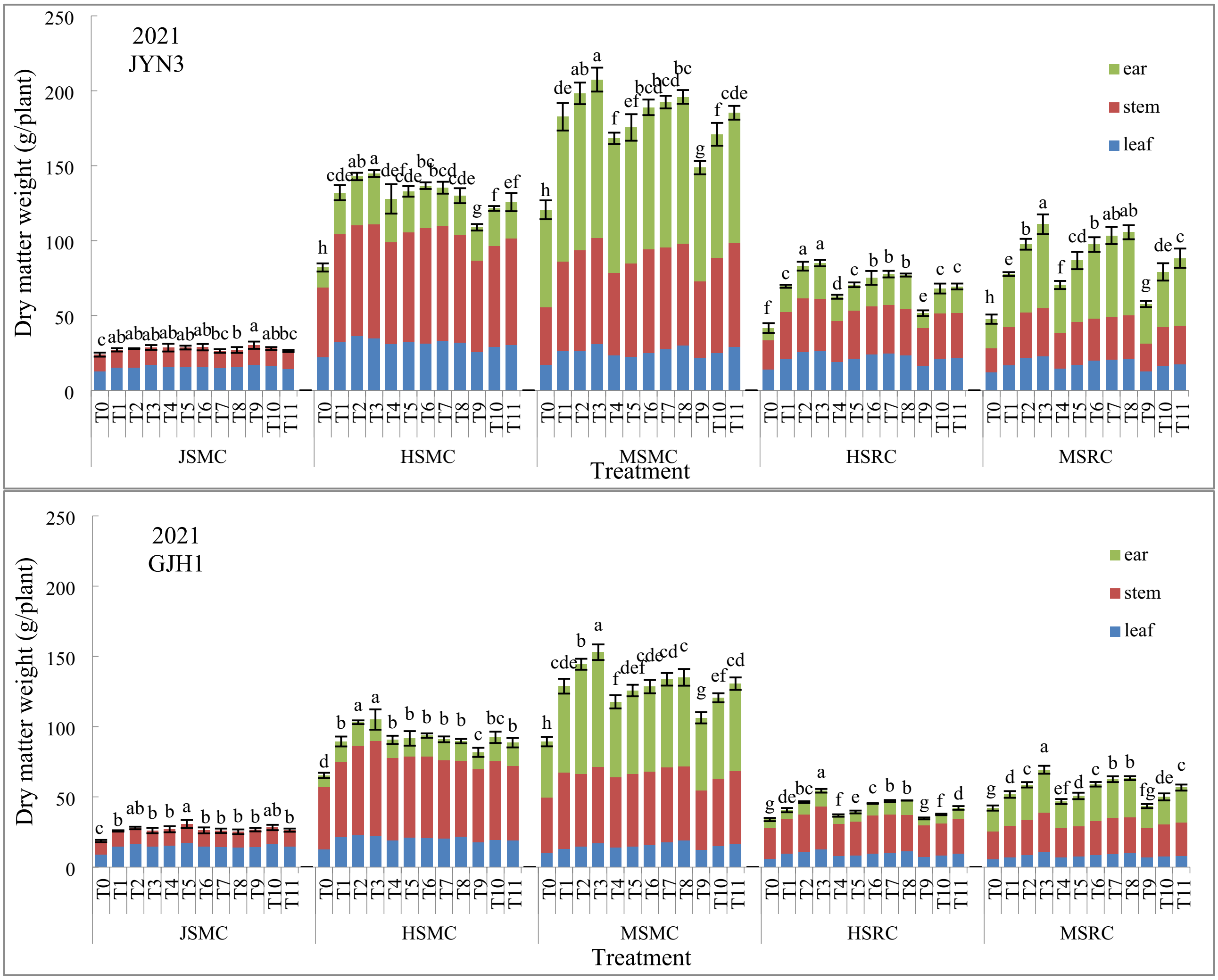

Fig. 1. (continued)

main and annual yields of GJH1 under T11 in 2020, the main, ratoon, and annual yields under T11 were about the same as or even higher than those under T1 and T4. These results suggested that lower rate of N fertilizer could produce grain yield as high as higher rate of N fertilizer by splitting application.

N use efficiencies

The RE_N, AE_N, and PFP_N in main season and whole year were significantly affected by cultivar (C), treatment (T), and their interactions (Table 7). JYN3 exhibited higher NUEs than GJH1. The average RE_N, AE_N, and PFP_N in main season across the three years were 35.33%, 12.93 kg kg^{-1}, and 44.38 kg kg^{-1} for JYN3, and 29.19%, 11.36 kg kg^{-1}, and 32.67 kg kg^{-1} for GJH1. The average RE_N, AE_N, and PFP_N in whole year were 70.76%, 24.27 kg kg^{-1}, and 69.22 kg kg^{-1} for JYN3, and 53.56%, 14.34 kg kg^{-1}, and 40.84 kg kg^{-1} for GJH1.

For both cultivars, decreasing the N application rate significantly increased RE_N, AE_N, and PFP_N in main season and whole year. For JYN3, the average RE_N, AE_N, and PFP_N in main season under 75 kg N ha^{-1} were higher with 51.96%, 100.12%, and 166.21% than those under 225 kg N ha^{-1}, and 29.51%, 49.73%, and 84.27% than those under 150 kg N ha^{-1}. The average RE_N, AE_N, and PFP_N in whole year under 75 kg N ha^{-1} were higher with 57.45%, 101.02%, and 160.72% than those under 225 kg N ha^{-1}, and 25.07%, 44.78%, and 79.10% than those under 150 kg N ha^{-1}. For GJH1, the average RE_N, AE_N, and PFP_N in main season under 75 kg N ha^{-1} were 43.46%, 84.01%, and 152.40% higher than those under 225 kg N ha^{-1}, and 22.35%, 43.44%, and 78.89% than those under 150 kg N ha^{-1}. The average RE_N, AE_N and PFP_N in whole year under 75 kg N ha^{-1} were 56.40%, 74.54% and 147.37% higher than those under 225 kg N ha^{-1}, and 22.35%, 37.14% and 76.10% higher than those under 150 kg N ha^{-1}.

NUEs for grain production of sorghum was increased by N splitting in both main season and whole year, and three-split pattern resulted in higher NUEs than two-split pattern. The RE_N, AE_N, and PFP_N reached the highest at T11, averaging 59.09%, 21.62 kg kg^{-1}, 74.84 kg kg^{-1} in main season, and 117.87%, 41.44 kg kg^{-1}, and 17.63 kg kg^{-1} in whole year for JYN3; and 46.80%, 18.65 kg kg^{-1}, and 54.71 kg kg^{-1} in main season, and 93.25%, 24.01 kg kg^{-1}, and 68.86 kg kg^{-1} in whole year for GJH1. The RE_N and AE_N in 225 kg N ha^{-1} splitting application (T2 and T3) were similar as, even higher than, those in 150 kg N ha^{-1} applied all as basal fertilizer (T4). Likewise, 150

Cultivar	Treatment	S-JSMC			JSMC-HSMC			HSMC-MSMC			MSMC-HSRC			HSRC-MSRC		
		2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021
JYN3	T0	2.94c	3.31f	3.41c	35.13g	33.81g	30.61f	10.64e	12.73g	11.57f	6.92g	6.36g	6.46f	1.72d	1.67f	1.38f
	N_0	**2.94C**	**3.31B**	**3.41B**	**35.13D**	**33.81D**	**30.61C**	**10.64B**	**12.73B**	**11.57B**	**6.92B**	**6.36C**	**6.46C**	**1.72B**	**1.67B**	**1.38B**
	T1	3.58b	3.98e	3.86ab	62.11b	58.54bc	55.18bc	16.13d	16.74ef	15.25d	11.20e	10.19cd	10.77c	2.22cd	2.13f	1.91f
	T2	3.82ab	4.38bcd	3.96ab	66.24a	63.32a	60.51a	17.55bcd	17.63e	16.67bcd	12.26cd	11.42b	12.85a	4.06b	5.48d	3.42de
	T3	3.91ab	4.50ab	4.09ab	66.63a	62.07ab	61.05a	20.57ab	19.68d	18.86ab	13.72a	12.43a	13.16a	5.07b	8.00ab	6.07ab
	N_{225}	**3.77B**	**4.29A**	**3.97A**	**64.99A**	**61.31A**	**58.91A**	**18.09A**	**18.02A**	**16.93A**	**12.39A**	**11.35A**	**12.26A**	**3.78A**	**5.20A**	**3.80A**
	T4	3.80ab	4.07cde	4.07ab	55.75d	53.54de	52.23cd	12.04e	13.97g	12.15f	10.09f	9.03ef	9.69d	2.87c	2.08f	1.52f
	T5	3.87ab	4.44bc	4.08ab	58.87cd	56.39cd	54.84bc	16.28cd	15.91f	12.83ef	11.26e	9.94de	10.92c	4.62b	5.42d	3.78de
	T6	3.88ab	4.15bcde	4.12ab	61.17bc	58.12c	56.72b	19.52ab	20.00d	15.72cd	12.24cd	10.79bcd	11.64b	6.50a	7.11bc	5.20bc
	T7	4.04a	4.13bcde	3.75bc	61.42bc	54.56cde	57.35b	19.64ab	22.91b	17.18bcd	13.18abc	11.51ab	12.02b	6.78a	6.45cd	6.02ab
	T8	3.95a	4.03de	3.85b	58.75cd	52.04e	54.18bc	22.14a	24.72a	19.83a	13.34ab	11.17bc	11.93b	6.97a	8.84a	6.68a
	N_{150}	**3.91AB**	**4.16A**	**3.97A**	**59.19B**	**54.93B**	**55.06A**	**17.92A**	**19.50A**	**15.54A**	**12.02A**	**10.49AB**	**11.24A**	**5.55A**	**5.98A**	**4.64A**
	T9	4.01a	4.83a	4.31a	48.79f	45.53f	41.40e	11.59e	13.69g	11.95f	9.85f	8.48f	7.99e	3.00c	2.03f	1.43f
	T10	4.17a	4.40bcd	3.98ab	52.67e	50.81e	49.23d	17.56bcd	16.06f	14.88de	11.58de	9.83de	10.54c	4.80b	4.31e	2.60ef
	T11	4.05a	4.14bcde	3.75bc	57.72d	50.75e	52.30cd	19.18abc	21.46c	17.92abc	12.49bcd	10.17cd	10.75c	6.33a	6.21cd	4.42cd
	N_{75}	**4.08A**	**4.45A**	**4.01A**	**53.06C**	**49.03C**	**47.64B**	**16.11A**	**17.07A**	**14.92A**	**11.31A**	**9.49B**	**9.76B**	**4.71A**	**4.18AB**	**2.81AB**
GJH1	T0	2.63d	3.00d	2.64c	24.42g	24.91e	24.58f	12.20c	9.30h	7.24e	5.58e	5.02f	5.24g	1.70d	1.82d	1.99f
	N_0	**2.63B**	**3.00B**	**2.64B**	**24.42C**	**24.91C**	**24.58D**	**12.20B**	**9.30C**	**7.24C**	**5.58B**	**5.02B**	**5.24C**	**1.70B**	**1.82B**	**1.99B**
	T1	3.55c	3.83c	3.67b	35.44de	35.02c	33.51bc	13.92bc	12.31ef	11.84c	6.72cd	6.15cd	6.28de	2.08d	3.26bc	2.61de
	T2	3.74abc	4.19ab	3.96ab	39.81ab	39.30a	39.61a	17.45a	13.73cd	12.41bc	6.91cd	6.87b	7.17bc	3.58abc	3.39bc	2.96bcd
	T3	3.77abc	4.27a	3.71b	41.46a	38.21ab	41.58a	18.21a	16.62a	14.41a	7.95ab	7.85a	8.41a	3.96a	4.00a	3.50a
	N_{225}	**3.69A**	**4.09A**	**3.78A**	**38.90A**	**37.51A**	**38.23A**	**16.53A**	**14.22A**	**12.89A**	**7.19A**	**6.96A**	**7.29A**	**3.21A**	**3.55A**	**3.02A**
	T4	3.52c	3.94bc	3.84b	34.72de	33.09c	33.49bc	11.76cd	10.04gh	8.14e	6.20de	5.46ef	5.69f	2.18d	3.10c	2.32ef
	T5	3.92a	4.05abc	4.36a	35.13de	33.26c	32.09cd	17.18a	12.73de	10.25d	6.75cd	6.31c	6.07e	2.61cd	3.33bc	2.67cde
	T6	3.75abc	4.27a	3.72b	36.00cde	34.11c	35.54b	18.88a	14.38bc	10.47d	7.50bc	6.87b	7.01c	3.73a	3.97a	3.20abc
	T7	3.71abc	3.99abc	3.68b	37.68bcd	35.21bc	34.26bc	18.68a	15.61ab	12.85bc	8.00ab	7.48a	7.29b	3.64abc	4.17a	3.63a
	T8	3.62bc	4.09abc	3.58b	38.75abc	35.70bc	33.96bc	19.13a	15.30ab	13.66ab	8.38a	7.51a	7.34b	3.66ab	4.28a	3.72a
	N_{150}	**3.70A**	**4.07A**	**3.83A**	**36.46A**	**34.27B**	**33.87B**	**17.12A**	**13.61AB**	**11.07AB**	**7.37A**	**6.72A**	**6.68AB**	**3.16A**	**3.77A**	**3.11A**
	T9	3.64abc	4.03abc	3.79b	30.66f	29.86d	28.96e	9.80cd	10.09gh	7.40e	6.18de	5.36ef	5.37g	1.91d	3.06c	2.06f
	T10	3.90ab	4.23ab	4.01ab	34.15e	34.37c	31.16d	12.21c	11.25fg	9.96d	6.47d	5.78de	5.81f	2.67bcd	3.25bc	2.92bcd
	T11	3.62bc	4.01abc	3.75	34.95de	35.67bc	32.75cd	14.78b	13.86cd	12.64bc	7.41bc	6.34c	6.52d	3.35abc	3.70ab	3.39ab
	N_{75}	**3.72A**	**4.09A**	**3.85A**	**33.26B**	**33.30B**	**30.96C**	**12.26B**	**11.73B**	**10.00B**	**6.69A**	**5.83B**	**5.90BC**	**2.64AB**	**3.34A**	**2.79A**
F value	Y	66.17**			72.59**			135.66**			108.07**			49.63**		
	C	62.55**			6983.08**			574.90**			5471.73**			321.02**		
	T	43.11**			222.27**			120.18**			183.44**			110.44**		
	Y×C	0.03			16.75**			49.07**			18.08**			13.44**		
	Y×T	1.59			1.42			3.88**			2.88**			2.02**		
	C×T	2.71**			29.89**			6.99**			28.74**			31.12**		
	Y×C×T	1.64*			1.54			1.96*			0.96			3.27**		

Table 3. The CGR (g m^{-2} d^{-1}) of different growth periods of two cultivars under different N treatments in 2019, 2020, and 2021. Within a column, means followed by different letters are significantly different according to Duncan's test (0.05). Lower-case and upper-case letters indicate comparisons among 12 N treatments and 4 N rates, respectively. N_0, N_{225}, N_{150}, and N_{75} represent N rates of 0, 225, 150, and 75 kg ha^{-1}. S: sowing; JSMC, jointing stage of MC; HSMC, heading stage of MC; MSMC, maturity stage of MC; HSRC, heading stage of RC; MSRC, maturity stage of RC. *, $P < 0.05$; **, $P < 0.01$. Significant values are in bold.

Cultivar	Treatment	JSMC			HSMC			MSMC			HSRC			MSRC		
		2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021
JYN3	T0	3.02de	3.32b	3.00g	4.96d	5.40d	4.42e	3.87e	3.90f	3.59g	2.84f	3.04g	3.07g	0.31j	0.58g	0.68f
	N_0	**3.02B**	**3.32B**	**3.00C**	**4.96C**	**5.40B**	**4.42C**	**3.87C**	**3.90C**	**3.59C**	**2.84B**	**3.04B**	**3.07B**	**0.31C**	**0.58B**	**0.68B**
	T1	4.16a	4.29a	4.04a	5.92ab	6.14c	5.71bc	5.37bc	5.59d	5.28d	3.97d	4.13d	3.99e	2.70f	2.51d	2.41c
	T2	3.91ab	4.09a	3.85ab	6.12a	7.43a	6.62a	5.71ab	6.38c	5.80b	4.63c	4.75c	4.61c	3.21de	2.84c	2.74b
	T3	4.00ab	4.04a	3.67bcd	6.10a	7.19ab	6.77a	5.95a	6.98a	6.30a	5.25ab	5.59a	5.27ab	3.79ab	3.83a	3.41a
	N_{225}	**4.03A**	**4.14A**	**3.86A**	**6.05A**	**6.92A**	**6.37A**	**5.68A**	**6.31A**	**5.79A**	**4.62A**	**4.82A**	**4.63A**	**3.23A**	**3.06A**	**2.85A**
	T4	3.61bc	4.01a	3.75bc	5.77ab	6.16c	5.54cd	5.36bc	5.10e	4.98e	3.74d	3.70ef	3.84e	2.44fg	2.16ef	2.02de
	T5	3.81ab	3.95a	3.48def	6.19a	7.19ab	6.70a	5.88a	6.51bc	5.85b	3.83d	4.54c	4.50cd	3.07e	2.42de	2.35c
	T6	3.21cde	3.52b	3.56cde	5.94ab	7.04b	6.51a	5.68ab	6.96a	6.29a	4.93bc	5.12b	5.08b	3.48cd	3.14b	2.94b
	T7	3.25cde	3.42b	3.34ef	6.01ab	7.26ab	6.83a	5.81a	6.99a	6.54a	5.45a	5.45ab	5.26ab	3.65bc	3.61a	3.36a
	T8	2.91e	3.40b	3.36ef	5.80ab	6.96b	6.52a	5.75ab	6.87ab	6.37a	5.58a	5.61a	5.45a	3.97a	3.82a	3.49a
	N_{150}	**3.36B**	**3.66B**	**3.50B**	**5.94A**	**6.92A**	**6.42A**	**5.70A**	**6.49A**	**6.01A**	**4.71A**	**4.88A**	**4.83A**	**3.32A**	**3.03A**	**2.83A**
	T9	3.60bc	3.95a	3.71bcd	5.35c	5.55d	5.31d	4.93d	4.77e	4.65f	3.27e	3.50f	3.52f	1.71i	1.96f	1.94e
	T10	3.35cd	3.54b	3.38ef	5.64bc	6.20c	5.78bc	5.24cd	5.59d	5.39cd	3.64de	4.01de	4.32d	2.10h	2.45de	2.29cd
	T11	3.05de	3.42b	3.25f	5.80ab	6.00c	5.91b	5.66abc	5.73d	5.59bc	4.83c	5.19b	4.75c	2.33gh	3.08bc	2.90b
	N_{75}	**3.33B**	**3.64B**	**3.45B**	**5.60B**	**5.92B**	**5.67B**	**5.28B**	**5.36B**	**5.21B**	**3.91A**	**4.23A**	**4.20A**	**2.05B**	**2.50A**	**2.37A**
GJH1	T0	2.41d	2.68e	2.57g	3.57f	4.01e	3.76f	2.94g	2.86f	2.67h	2.15d	2.47h	2.71i	0.00h	0.00i	0.00f
	N_0	**2.41C**	**2.68C**	**2.57B**	**3.57C**	**4.01D**	**3.76D**	**2.94C**	**2.86D**	**2.67C**	**2.15B**	**2.47B**	**2.71D**	**0.00C**	**0.00C**	**0.00C**
	T1	3.73a	4.02ab	3.88a	4.62cd	5.28b	5.05b	4.03de	4.42b	4.10cde	2.90c	3.17f	4.12def	0.81e	0.58f	0.59c
	T2	3.75a	3.98ab	3.57bc	4.98ab	5.51ab	5.36a	4.58b	4.93a	4.52b	3.25b	3.59de	4.66ab	1.00cd	0.87c	0.74b
	T3	3.35b	3.56c	3.37cd	5.17a	5.65a	5.15ab	4.95a	5.02a	4.83a	3.84a	4.23ab	4.80a	1.20a	1.09a	0.98a
	N_{225}	**3.61A**	**3.85A**	**3.61A**	**4.92A**	**5.48A**	**5.18A**	**4.52A**	**4.79A**	**4.48A**	**3.33A**	**3.66A**	**4.53A**	**1.00A**	**0.85A**	**0.77A**
	T4	3.53ab	4.12a	3.86a	4.11e	4.56c	4.47cd	3.75e	4.04d	3.75fg	2.66c	3.14f	3.80g	0.78ef	0.43g	0.45e
	T5	3.40b	3.82b	3.53bc	4.67bc	4.75c	4.68c	4.48bc	4.00d	3.90ef	2.90c	3.39ef	3.99efg	0.90de	0.74e	0.47de
	T6	3.03c	3.54c	3.15ef	4.48cd	4.64c	4.31de	4.26cd	4.39bc	4.11cde	3.35b	4.01bc	4.19de	0.99cd	0.79d	0.56cd
	T7	3.06c	3.18d	3.02ef	4.65c	4.71c	4.48cd	4.59b	4.41b	4.25bcd	3.83a	4.11ab	4.38cd	1.05bc	0.85c	0.76b
	T8	2.96c	3.07d	3.07ef	4.58cd	4.63c	4.39de	4.54bc	4.48b	4.32bc	3.91a	4.37a	4.54bc	1.16ab	0.97b	0.83b
	N_{150}	**3.20B**	**3.55AB**	**3.33A**	**4.50B**	**4.66B**	**4.47B**	**4.32A**	**4.26B**	**4.07B**	**3.33A**	**3.81A**	**4.18B**	**0.98A**	**0.76A**	**0.61A**
	T9	3.41b	3.54c	3.59b	4.02e	4.52cd	4.13e	3.28f	3.65e	3.51g	2.39d	2.76g	3.33h	0.45g	0.36h	0.00f
	T10	3.02c	3.41c	3.23de	4.33cde	4.46cd	4.20de	3.99de	4.04d	3.88ef	2.73c	3.37ef	3.85g	0.67f	0.43hg	0.44e
	T11	2.94c	3.44c	2.98f	4.30de	4.24de	4.22de	4.17d	4.13cd	3.95def	3.35b	3.78cd	3.88fg	0.85e	0.57f	0.49cde
	N_{75}	**3.13B**	**3.46B**	**3.27A**	**4.21B**	**4.41C**	**4.18C**	**3.81B**	**3.94C**	**3.78B**	**2.82A**	**3.30A**	**3.68C**	**0.66B**	**0.45B**	**0.31B**
F value	Y	57.25**			144.92**			62.96**			168.34**			68.42**		
	C	108.82**			3743.32**			3809.17**			1434.66**			13,876.28**		
	T	83.08**			121.40**			234.21**			286.10**			417.88**		
	Y×C	0.72			26.41**			26.68**			94.38**			7.11**		
	Y×T	1.62*			2.81**			3.39**			3.75**			6.65**		
	C×T	4.12**			18.30**			17.23**			13.30**			108.10**		
	Y×C×T	1.36			5.40**			5.67**			2.21**			5.50**		

Table 4. The LAI of different growth stages of two cultivars under different N treatments in 2019, 2020, and 2021. Within a column, means followed by different letters are significantly different according to Duncan's test (0.05). Lower-case and upper-case letters indicate comparisons among 12 N treatments and 4 N rates, respectively. N_0, N_{225}, N_{150}, and N_{75} represent N rates of 0, 225, 150, and 75 kg ha^{-1}. JSMC, jointing stage of MC; HSMC, heading stage of MC; MSMC, maturity stage of MC; HSRC, heading stage of RC; MSRC, maturity stage of RC. *, $P<0.05$; **, $P<0.01$. Significant values are in bold.

kg N ha^{-1} applied at two and three splits (T5, T6, T7, and T8) caused similar or higher RE_N and AE_N compared with the entire 75 kg N ha^{-1} applied once before planting (T9).

Discussion

Growth parameters

Dry matter accumulation serves as the fundamental basis of yield formation, particularly post-anthesis dry matter accumulation, which contributes over two thirds to grain yield[23]. The potential grain yield is closely associated with the CGR of late reproductive period[24]. LAI, a parameter characterizing crop canopy structure, not only estimates current dry matter production but also predicts subsequent growth and yields[25]. PP, the product of green leaf area per unit area and photosynthetic time, was regarded as an important indicator of population

Cultivar	Treatment	S-JSMC			JSMC- HSMC			HSMC-MSMC			MSMC-HSRC			HSRC-MSRC		
		2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021
JYN3	T0	114.86de	114.68b	111.14g	71.80d	87.24f	74.21e	145.65e	162.77d	140.11g	218.01h	242.86e	226.48f	78.71h	63.35f	84.40h
	N_0	**114.86B**	**114.68B**	**111.14C**	**71.80C**	**87.24C**	**74.21C**	**145.65C**	**162.77C**	**140.11C**	**218.01C**	**242.86C**	**226.48C**	**78.71C**	**63.35B**	**84.40B**
	T1	158.24a	147.91a	149.62a	90.79b	104.25cd	97.51bc	186.40abc	205.16b	192.19de	303.64ef	340.19c	315.10cd	166.61de	116.24d	143.94ef
	T2	148.62ab	141.02a	142.62ab	90.29ab	115.17a	104.73a	195.22ab	241.61a	217.38c	336.15cd	389.38b	354.21b	196.19c	132.74c	165.40cd
	T3	152.13ab	139.50a	135.98bcd	90.91a	112.34ab	104.48a	198.86a	247.91a	228.85ab	364.09ab	439.80a	393.54a	225.88ab	164.90a	195.34a
	N_{225}	**153.00A**	**142.81A**	**142.74A**	**90.66A**	**110.59A**	**102.24A**	**193.49A**	**231.56A**	**212.81A**	**334.63A**	**389.79A**	**354.28AB**	**196.23A**	**137.96A**	**168.23A**
	T4	137.04bc	138.43a	138.78bc	84.43abc	101.74cd	92.94cd	183.76bc	197.10b	184.23e	295.93ef	308.04d	300.15d	154.59ef	102.52e	131.98fg
	T5	144.72ab	136.16a	128.80def	90.01ab	111.40ab	101.78ab	199.27a	239.82a	219.57bc	315.83de	386.74b	351.83b	172.55d	121.83d	154.09de
	T6	122.07cde	121.41b	131.65cde	82.37c	105.58bc	100.69ab	191.75abc	244.92a	224.01abc	344.86abc	422.75a	386.57a	210.36bc	144.62b	180.45b
	T7	123.39cde	117.92b	123.66ef	83.36bc	106.82bc	101.71ab	195.05ab	249.44a	234.03a	365.87ab	435.30a	401.40a	227.59a	158.51a	193.99a
	T8	110.64e	117.15b	124.20ef	78.40c	103.56cd	98.76b	190.57abc	242.11a	225.64abc	368.32a	436.92a	402.09a	238.71a	164.92a	201.20a
	N_{150}	**127.57B**	**126.21B**	**129.42B**	**83.71B**	**105.82A**	**99.18A**	**192.08A**	**234.68A**	**217.50A**	**338.16A**	**397.95A**	**368.41A**	**200.76A**	**138.48A**	**172.34A**
	T9	136.78bc	136.28a	137.36bcd	80.51c	94.96e	90.25d	169.52d	180.48c	174.25f	266.48g	289.29d	277.78e	124.51g	95.46e	122.87g
	T10	127.25cd	122.00b	125.14ef	80.94c	97.37de	91.60d	179.68cd	206.37b	195.39d	288.76fg	336.07c	330.06c	143.49f	113.07d	148.69e
	T11	115.88de	118.09b	120.44f	79.61c	94.26e	91.64d	188.95abc	205.40b	201.28d	340.77bc	382.25b	351.81b	178.91d	144.65b	172.25bc
	N_{75}	**126.64B**	**125.46B**	**127.64B**	**80.35B**	**95.53B**	**91.16B**	**179.38B**	**197.42B**	**190.31B**	**298.67B**	**335.87B**	**319.88B**	**148.97B**	**117.73A**	**147.94A**
GJH1	T0	91.71d	92.34e	95.15g	53.84e	66.90e	63.29e	107.46g	120.27f	112.55f	165.64f	186.67g	183.08i	53.82e	43.30g	60.98f
	N_0	**91.71C**	**92.34C**	**95.15B**	**53.84C**	**66.90C**	**63.29C**	**107.46C**	**120.27D**	**112.55D**	**165.64C**	**186.67C**	**183.08D**	**53.82C**	**43.30C**	**60.98C**
	T1	141.84a	138.68ab	143.71a	75.17ab	93.03a	89.30a	142.67cd	169.80b	159.98b	225.18cd	265.49de	279.31def	92.87c	65.55e	105.94c
	T2	142.33a	137.45ab	132.08bc	78.56a	94.97a	89.29a	157.78ab	182.78a	172.98a	254.27b	298.21b	312.28ab	106.14b	78.12c	121.43b
	T3	127.31b	122.85c	124.77cd	76.68ab	92.08a	85.20ab	166.98a	186.65a	174.53a	285.66a	323.85a	327.26a	125.90a	93.26a	129.96a
	N_{225}	**137.16A**	**132.99A**	**133.52A**	**76.80A**	**93.36A**	**87.93A**	**155.81A**	**179.74A**	**169.16A**	**255.04A**	**295.85A**	**306.28A**	**108.30A**	**78.98A**	**119.11A**
	T4	134.15ab	142.11a	142.82a	68.74cd	86.84b	83.32b	129.65ef	150.51cde	143.88cd	208.22d	251.27e	256.71g	85.84c	62.61e	95.60d
	T5	129.31b	131.63b	130.57bc	72.68bc	85.67bc	82.10b	150.95bc	153.18cde	150.23bcd	239.61bc	258.75de	268.28efg	94.99c	72.28d	100.26cd
	T6	115.27c	122.12c	116.59ef	67.64cd	81.80cd	74.63cd	144.26cd	158.08cd	147.38cd	247.44b	294.24bc	282.37de	108.62b	84.03b	107.01c
	T7	116.32c	109.78d	111.68ef	69.37cd	78.91d	74.95cd	152.49bc	159.64c	152.63bc	273.72a	298.40b	293.16cd	121.97a	86.84b	115.47b
	T8	112.60c	106.00d	113.50ef	67.87cd	77.04d	74.59cd	150.41bc	159.47c	152.40bc	274.50a	309.62ab	300.97bc	126.78a	93.34a	120.70b
	N_{150}	**121.53B**	**122.33AB**	**123.03A**	**69.26B**	**82.05B**	**77.92B**	**145.55A**	**156.18B**	**149.31B**	**248.70A**	**282.46A**	**280.30B**	**107.64A**	**79.82A**	**107.81A**
	T9	129.67b	122.15c	132.99b	66.92d	80.57d	77.22c	120.52f	142.84e	133.69e	184.40e	224.07f	232.71h	71.01d	54.55f	75.00e
	T10	114.92c	117.65c	119.48de	66.15d	78.75d	74.32cd	137.23de	148.76de	141.42de	218.36d	259.30de	262.62fg	84.96c	66.60e	96.35d
	T11	111.82c	118.74c	110.09f	65.14d	76.80d	71.98d	139.61d	146.53e	143.04cde	244.33b	276.91cd	266.08efg	105.00b	76.06cd	98.20d
	N_{75}	**118.80B**	**119.51B**	**120.85A**	**66.07B**	**78.71B**	**74.51B**	**132.45B**	**146.04C**	**139.38C**	**215.70B**	**253.43B**	**253.80C**	**86.99B**	**65.74B**	**89.85B**
F value	Y	1.92			486.30**			276.27**			251.87**			714.65**		
	C	108.58**			1572.99**			4085.09**			2671.45**			5356.82**		
	T	82.38**			86.87**			180.42**			267.49**			354.09**		
	Y×C	1.19			13.40**			40.47**			25.64**			84.92**		
	Y×T	1.73*			1.13			3.32**			2.12**			6.88**		
	C×T	4.05**			5.33**			19.12**			14.85**			41.13**		
	Y×C×T	1.35			2.42**			5.52**			2.42**			2.98**		

Table 5. The PP ($m^2\ d^{-1}\ m^{-2}$) of different growth stages of two cultivars under different N treatments in 2019, 2020, and 2021. Within a column, means followed by different letters are significantly different according to Duncan's test (0.05). Lower-case and upper-case letters indicate comparisons among 12 N treatments and 4 N rates, respectively. N_0, N_{225}, N_{150}, and N_{75} represent N rates of 0, 225, 150, and 75 kg ha^{-1}. S: sowing; JSMC, jointing stage of MC; HSMC, heading stage of MC; MSMC, maturity stage of MC; HSRC, heading stage of RC; MSRC, maturity stage of RC. *, $P<0.05$; **, $P<0.01$. Significant values are in bold.

Cultivar	Treatment	Main yield (kg ha^{-1})			Ratoon yield (kg ha^{-1})			Annual yield (kg ha^{-1})		
		2019	2020	2021	2019	2020	2021	2019	2020	2021
JYN3	T0	3881.05e	4143.85g	3948.19h	1840.25e	1626.90h	1703.45i	5722.97i	5770.75 k	5651.64f
	N0	**3881.05C**	**4143.85D**	**3948.19C**	**1840.25B**	**1626.90C**	**1703.45C**	**5722.97C**	**5770.75C**	**5651.64C**
	T1	5502.90b	5903.00cd	5729.20cd	3019.20c	3018.45e	3140.50e	8523.77ef	8921.45fg	8869.70c
	T2	5966.50a	6393.20a	6192.95a	3374.30b	3477.55bc	3444.05cd	9340.80bc	9870.75ab	9637.00b
	T3	5983.60a	6265.10ab	6347.15a	3806.00a	3766.45a	3944.95a	9789.60a	10,031.55a	10,292.10a
	N225	**5817.67A**	**6187.10A**	**6089.77A**	**3399.83A**	**3420.82A**	**3509.83A**	**9218.06A**	**9607.92A**	**9599.60A**
	T4	5280.55bc	5455.50e	5354.20f	2778.35d	2741.40f	2670.20g	8058.90g	8196.90i	8024.40d
	T5	5791.35a	6064.95bc	5735.20cd	3279.55b	3270.30d	3333.30d	9070.90cd	9335.25de	9068.50c
	T6	5823.00a	5842.05cd	5906.45bc	3395.65b	3310.40cd	3555.15bc	9218.65c	9152.45ef	9461.60b
	T7	5918.20a	6073.50bc	5986.15b	3722.60a	3579.55ab	3695.25b	9640.80ab	9653.05bc	9681.40b
	T8	5902.70a	5801.10d	6202.90a	3856.15a	3717.80a	3885.70a	9758.85a	9518.90cd	10,088.60a
	N150	**5743.16A**	**5847.42B**	**5836.98A**	**3406.46A**	**3323.89A**	**3427.92A**	**9149.62A**	**9171.31A**	**9264.90A**
	T9	4932.75d	5069.10f	4921.40g	2570.25d	2432.45g	2426.50h	7503.00h	7501.55j	7347.90e
	T10	5227.50c	5682.30d	5498.60ef	3174.50bc	2843.65ef	2862.85f	8402.00fg	8525.95h	8361.45d
	T11	5447.30bc	5739.00d	5651.60de	3342.25b	3016.05e	3270.60de	8789.55de	8755.05gh	8922.20c
	N75	**5202.52B**	**5496.80C**	**5357.20B**	**3029.00A**	**2764.05B**	**2853.32B**	**8231.52B**	**8260.85B**	**8210.52B**
GJH1	T0	2705.55g	2810.90g	2597.25i	694.85f	651.80g	601.10g	3400.40h	3462.70f	3198.35i
	N0	**2705.55C**	**2810.90C**	**2597.25D**	**694.85C**	**651.80C**	**601.10C**	**3400.40C**	**3462.70C**	**3198.35C**
	T1	4223.50c	4423.65d	3982.25de	1013.50cd	1034.25de	963.95d	5237.00d	5457.90c	4946.20e
	T2	4774.65b	4835.60b	4245.55b	1235.15b	1183.45c	1168.50bc	6009.80b	6019.05b	5414.05bc
	T3	5017.15a	5046.10a	4738.30a	1456.85a	1423.65a	1379.50a	6474.00a	6469.75a	6117.80a
	N225	**4671.77A**	**4768.45A**	**4322.03A**	**1235.17A**	**1213.78A**	**1170.65A**	**5906.93A**	**5982.23A**	**5492.68A**
	T4	3681.40e	4020.95e	3472.10g	901.25de	882.10f	845.70e	4582.65f	4903.05d	4317.80g
	T5	4205.20c	4603.45cd	3885.35e	1043.90c	984.90ef	1091.55c	5249.10d	5588.35c	4976.90e
	T6	4656.90b	4772.25bc	4069.30cd	1017.40cd	1121.35cd	1178.45bc	5674.30c	5893.60b	5247.75cd
	T7	4604.05b	4800.80bc	4246.95b	1058.55c	1200.50bc	1270.50b	5662.60c	6001.30b	5517.45b
	T8	4677.10b	4835.55b	4191.10bc	1341.20ab	1313.75ab	1416.90a	6018.30b	6149.30b	5608.00b
	N150	**4364.93A**	**4606.60A**	**3972.96B**	**1072.46AB**	**1100.52A**	**1160.62A**	**5437.39AB**	**5707.12A**	**5133.58A**
	T9	3510.95f	3734.55f	3292.80h	805.40ef	763.75g	717.55f	4316.35g	4498.30e	4010.35h
	T10	4027.40d	4147.45e	3711.85f	964.30cd	900.55f	904.80de	4991.70e	5048.00d	4616.65f
	T11	4261.55c	4160.00e	3888.70e	1064.60c	952.00ef	1166.80bc	5326.15d	5112.00d	5055.50de
	N75	**3933.30B**	**4014.00B**	**3631.12C**	**944.77B**	**872.10B**	**929.72B**	**4878.07B**	**4886.10B**	**4560.83B**
F value	Y	110.71**			10.40**			32.63**		
	C	8442.70**			26,564.36**			24,216.31**		
	T	484.77**			347.61**			656.60**		
	Y × C	89.80**			5.15**			46.93**		
	Y × T	2.17**			2.69**			1.91*		
	C × T	4.82**			80.37**			29.39**		
	Y × C × T	3.14**			1.37			2.10**		

Table 6. The main, ratoon, and annual yields of two cultivars under different N treatments in 2019, 2020, and 2021. Within a column, means followed by different letters are significantly different according to Duncan's test (0.05). Lower-case and upper-case letters indicate comparisons among 12 N treatments and 4 N rates, respectively. N_0, N_{225}, N_{150}, and N_{75} represent N rates of 0, 225, 150, and 75 kg ha^{-1}. *, $P < 0.05$; **, $P < 0.01$. Significant values are in bold.

photosynthetic capacity[26]. N fertilizer promotes the growth of sorghum, as evidenced by increased LAI, total dry matter per plant, and CGR with increasing N rates[27]. In this study, all growth parameters including dry matter weight, CGR, LAI, and PP exhibited an increase with increasing N rates.

It worthy mentioned that, splitting the application of N on sweet potato resulted in increased chlorophyll content, leaf area duration (LAD), net assimilation rate (NAR), and biomass accumulated during the intermediate and final growth phases compared to a single application as basal fertilizer[28]. Xu et al.[29] reported that postponing topdressing N fertilizer treatments enhanced the maximum average CGR of wheat/maize intercropping. Shifting appropriate nitrogen application amounts to the late growth stage of MC effectively improved LAI, SPAD, and total dry matter accumulation in both MC and its ratoon rice crop[30]. Which with the same to our study that, splitting N application significantly promoted dry matter accumulation, CGR, LAI, and PP at almost all growth stages for both main and ratoon sorghum. Several studies on rice suggested that properly postponing

Cultivar	Treatment	The main season									The whole year								
		RE_N (%)			AE_N (kg kg^{-1})			PFP_N (kg kg^{-1})			RE_N (%)			AE_N (kg kg^{-1})			PFP_N (kg kg^{-1})		
		2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021
JYN3	T1	18.72g	18.09e	20.27g	7.21e	7.82f	7.92g	24.46f	26.24f	25.46h	35.76i	33.78h	35.95h	12.45g	14.00f	14.30h	37.88h	39.65h	39.42i
	T2	32.84e	31.24d	29.87e	9.27d	10.00cdef	9.98f	26.52f	28.41e	27.52g	59.65g	60.03f	57.69e	16.09f	18.22e	17.71g	41.51g	43.87g	42.83h
	T3	37.31c	37.34c	34.80d	9.34d	9.43def	10.66ef	26.59f	27.84ef	28.21g	70.15ef	74.75e	71.94d	18.08f	18.94e	20.62f	43.51g	44.58g	45.74h
	N_{225}	**29.63B**	**28.89B**	**28.31B**	**8.61C**	**9.08B**	**9.52C**	**25.86C**	**27.50C**	**27.07C**	**55.19B**	**56.19B**	**55.19B**	**15.54C**	**17.05C**	**17.55C**	**40.97C**	**42.70C**	**42.66C**
	T4	21.00g	20.99e	23.87f	9.33d	8.74ef	9.37fg	35.20e	36.37d	35.69f	42.60h	37.32h	42.17g	15.58fg	16.17ef	15.82gh	53.73f	54.65f	53.50g
	T5	33.97de	31.48d	30.22e	12.74c	12.81b	11.91de	38.61d	40.43c	38.23e	66.82f	62.81f	56.61e	22.33e	23.76cd	22.78e	60.47e	62.24de	60.46f
	T6	34.60cde	36.77c	36.02d	12.95c	11.32bcd	13.06d	38.82d	38.95c	39.38e	74.56de	73.61e	74.95d	23.32de	22.54d	25.40d	61.46de	61.02d	63.08ef
	T7	36.16cd	40.34c	39.92c	13.58c	12.86b	13.59cd	39.45d	40.49c	39.91de	77.84cd	85.48d	85.78c	26.13cd	25.88c	26.87d	64.27d	64.35d	64.54de
	T8	36.12cd	45.57b	42.38c	13.48c	11.05bcde	15.03c	39.35d	38.67c	41.35d	81.26bc	91.88c	94.81b	26.92c	24.99cd	29.58c	65.06d	63.46d	67.26d
	N_{150}	**32.37B**	**35.03B**	**34.48B**	**12.41B**	**11.36B**	**12.59B**	**38.29B**	**38.98B**	**38.91B**	**68.62AB**	**70.22B**	**70.86AB**	**22.86B**	**22.67B**	**24.09B**	**61.00B**	**61.14B**	**61.77B**
	T9	24.23f	27.95d	26.55f	14.02c	12.34bc	12.98d	65.77c	67.59b	65.62c	56.82g	52.36g	46.96f	23.76cde	23.08cd	22.62e	100.04c	100.02c	97.97c
	T10	44.86b	49.34b	45.63b	17.95b	20.51a	20.67b	69.70b	75.76a	73.31b	85.19b	100.83b	91.01b	35.74b	36.74b	36.13b	112.03b	113.68b	111.49b
	T11	53.14a	59.72a	64.42a	20.88a	21.27a	22.71a	72.63a	76.52a	75.35a	105.65a	123.96a	124.01a	40.91a	39.79a	43.61a	117.19a	116.73a	118.96a
	N_{75}	**40.74A**	**45.67A**	**45.53A**	**17.62A**	**18.04A**	**18.79A**	**69.37A**	**73.29A**	**71.43A**	**82.55A**	**92.38A**	**87.32A**	**33.47A**	**33.20A**	**34.12A**	**109.75A**	**110.14A**	**109.47A**
GJH1	T1	12.91f	15.53h	12.94g	6.75e	7.17f	6.16h	18.77h	19.66f	17.70j	24.47e	25.21f	21.59f	8.16g	8.87e	7.63g	23.28i	24.26g	21.98 k
	T2	22.26e	27.85f	26.78e	9.20d	9.00de	7.33g	21.22g	21.49e	18.87i	37.90d	43.81e	44.12d	11.60f	11.36d	9.58f	26.71h	26.75f	23.93j
	T3	26.43d	35.30de	39.51c	10.27d	9.93d	9.52ef	22.30g	22.43e	21.06h	50.85c	62.18d	66.40b	13.66e	13.36c	13.02d	28.77g	28.75f	27.37i
	N_{225}	**20.53B**	**26.23A**	**26.41A**	**8.74B**	**8.70C**	**7.67B**	**20.76C**	**21.19C**	**19.21C**	**37.74B**	**43.73B**	**44.04A**	**11.14B**	**11.20C**	**10.08B**	**26.25C**	**26.59C**	**24.43C**
	T4	13.34f	16.41h	14.66g	6.51e	8.07ef	5.83h	24.54f	26.81d	23.15g	23.33e	28.79f	24.14f	7.88g	9.60de	7.73g	30.55g	32.69e	29.25h
	T5	26.28d	28.39f	28.06e	10.00d	11.95c	8.59f	28.03e	30.69c	25.90f	41.51d	46.40e	46.79d	12.32f	14.17c	11.66e	34.99f	37.26d	33.18g
	T6	27.65cd	34.05e	32.19d	13.01c	13.08bc	9.81de	31.05d	31.82c	27.13e	49.50c	63.35d	57.83c	15.16d	16.21b	13.26d	37.83e	39.29cd	34.79f
	T7	26.66d	37.09cd	38.40c	12.66c	13.27bc	11.00c	30.69d	32.01c	28.31d	58.85b	72.69c	71.69b	15.08d	16.92b	15.16c	37.75e	40.01c	36.68e
	T8	31.87b	40.78b	33.16d	13.14c	13.50b	10.63cd	31.18d	32.24c	27.94de	65.25b	81.74b	70.30b	17.45c	17.91b	15.76c	40.12d	41.00c	37.29d
	N_{150}	**25.16AB**	**31.34A**	**29.29A**	**11.06B**	**11.97B**	**9.17B**	**29.10B**	**30.71B**	**26.49B**	**47.69AB**	**58.59AB**	**54.15A**	**13.58B**	**14.96B**	**12.71B**	**36.25B**	**38.05B**	**34.24B**
	T9	21.26e	20.53g	19.49f	10.74d	12.32bc	9.27ef	46.81c	49.79b	43.90c	37.67d	39.67e	30.49e	12.21f	13.81c	10.83ef	57.55c	59.98b	53.87c
	T10	30.56bc	38.90bc	43.76b	17.62b	17.82a	14.86b	53.70b	55.30a	49.49b	61.16b	69.50cd	70.67b	21.22b	21.14a	18.51b	66.56b	67.31a	61.56b
	T11	36.44a	51.01a	52.94a	20.75a	17.99a	17.22a	56.82a	55.47a	51.85a	82.17a	97.65a	99.93a	25.68a	21.99a	24.36a	71.02a	68.16a	67.41a
	N_{75}	**29.42A**	**36.81A**	**38.73A**	**16.37A**	**16.04A**	**13.78A**	**52.44A**	**53.52A**	**48.41A**	**60.33A**	**68.94A**	**67.03A**	**19.70A**	**18.98A**	**17.90A**	**65.04A**	**65.15A**	**60.94A**
F value	Y	104.75**			8.54**			92.52**			52.21**			1.41			22.88**		
	C	507.30**			121.54**			7231.36**			1105.32**			2926.51**			19,458.15**		
	T	556.61**			284.92**			4626.00**			707.10**			468.42**			4149.49**		
	Y×C	24.86**			43.52**			75.64**			9.58**			21.63**			31.27**		
	Y×T	9.83**			1.66*			3.03**			7.70**			2.38**			2.24**		
	C×T	8.22**			4.91**			146.30**			6.26**			40.46**			318.57**		
	Y×C×T	5.38**			2.62**			3.44**			2.78**			1.42			1.16		

Table 7. The NUEs in main season and whole year of two cultivars under different N treatments in 2019, 2020, and 2021. Within a column, means followed by different letters are significantly different according to Duncan's test (0.05). Lower-case and upper-case letters indicate comparisons among 12 N treatments and 4 N rates, respectively. N_0, N_{225}, N_{150}, and N_{75} represent N rates of 0, 225, 150, and 75 kg ha^{-1}. *, $P < 0.05$; **, $P < 0.01$. Significant values are in bold.

N application in MC was beneficial for increasing dry matter accumulation, LAI, NAR, CGR, and root activity; thus enhancing yield in MC and RC[31–33].

Yield

Nitrogen is the most crucial nutrient for sorghum growth, yet it still limits its yield[34]. Inappropriate fertilizer amounts during cultivation can lead to reduced plant performance and fertilizer use efficiency[35]. Generally, sorghum grain yield initially increases with increasing N rates, but eventually plateaus or even declines[36]. In the present study, the grain yield of sorghum increased with N rate increasing; however, there was no significant difference between the 225 and 150 kg N ha^{-1} in terms of main, ratoon, and annual yields. It seems like that the increase of grain yield was due to the positive impact of nitrogen on photosynthesis, CGR, LAI, and LAD[37]. For ratoon rice, N_{bud} had a greater impact on grain yield due to increased LAI, total dry weight, and harvest index (HI)[7]. However, excessive N application rates can hinder canopy photosynthesis due to mutual shading and poor carbohydrate translocation from straw to grain, thus causing lower grain filling percentage and a lower HI[38,39].

Previous experiments on the N management mostly focused on single crop sorghum[40,41], thus the optimal N rate for high yield suggested by these studies might be surplus and could be used by ratoon sorghum[42]. Wang et al.[43] reported a high risk of residue N when the application rate exceeded 150 kg ha^{-1}. Our study demonstrated that grain yield could be achieved from ratoon crops without additional N application, confirming the presence of residual N after the harvest of the current crop. However, further research is needed to determine the precise amount of residual N in soil and to identify the optimal N application rate for ratoon crops.

Another approach to increase crop yield is by splitting the application of N to synchronize fertilizer nutrient supply with crop nutrient demands[44]. In this study, under a given N rate, splitting applications resulted in higher grain yields compared to single-dose application; furthermore, three-split applications yielded more than two-split applications. Overall, split-application of N demonstrated better performance than applying the entire amount at once[45–47]. Melaku et al.[48] reported that splitting application of 41, 64, and 87 kg N ha^{-1}, half at planting and half at knee height stage, led to 19%, 15% and 18% increase in sorghum grain yield compared to a single-dose application, respectively. Moreover, multi-split-N fertigation showed potential for increasing maize grain yield than two-split application under same nitrogen rate[49]. Three-split application increased maize yield by 10.7% compared with single application, while two-split application increased by 8.3%[50].

The effect of splitting N application on the increase in yield of ratoon sorghum was found to be more significant than that of main sorghum. This is attributed to the fact that a single basal application led to greater N loss before jointing and a lower residual N amount in the soil profile layer[51]. The application of N in three splits resulted in a notable increase in residual N in the 0–1 m soil[52], as well as an enhanced recovery of residual fertilizer N during the second cropping season, when compared to applying N with two splits[53]. Furthermore, it is suggested that an increased ratio of heading-N fertilizer could significantly increase the ratoon yield due to a relatively delay late-season leaf senescence, which was positively correlated with regenerative ability of RC[13].

NUEs

Nitrogen use efficiency (NUE) is defined as the percentage of fertilizer N which is recovered or utilized by a fertilized crop[54]. The selection of high-efficiency crop varieties and optimization of agronomic management practices to increase NUEs represent an effective strategy for increasing crop yield and reducing associated environmental costs[55]. Variability in NUEs has been demonstrated in sorghum, with high NUEs varieties exhibiting greater plant dry matter yields and grain yields[56,57]. In the present study, JYN3, which displayed higher grain yield and total dry weight, demonstrated higher NUEs compared with GJH1. These findings are consistent with those of Buah et al.[58], who assessed the agronomic responsiveness of 13 sorghum genotypes differing in NUE to three N rates.

It is widely acknowledged that NUEs is significantly influenced by N management practices, particularly rates, timing, resources, and placement of N fertilizer application[59–62]. The NUEs of grain sorghum decreased with increased N rate[63,64]. In this study, with the N rate increased from 75 to 225 kg ha^{-1}, RE_N, AE_N, and PFP_N in main season and whole year decreased for both sorghum cultivars. Another effective approach to increase NUEs would be to split-apply the fertilizer N[65]. Plants require less N fertilizer during the seedling growth stage, therefore, reducing the N application rate at this stage and postponing the N fertilization period could effectively reduce N application while increase yield and NUEs[66]. Reducing tillering-N ratios and increasing panicle-N ratios promoted aboveground N accumulation and plant N accumulation derived from fertilizer[67]. In our study, split-N application significantly improved NUEs in both main season and whole year compared to the basal application of all N, and three-split pattern, i.e., increasing the ratio of heading-N fertilizer, resulted in higher NUEs than two-split pattern. These findings are consistent with previous results obtained from studies on cotton, wheat, maize, rice and other crops[48,68–71].

In the present study, splitting the application into three separate doses at N rates of 150, and 75 kg ha^{-1} resulted in yields equivalent to or higher than those achieved with a single application at a rate of 225 kg N ha^{-1}. Furthermore, the RE_N and AE_N observed in two- and three-splitting application at higher N rates were similar to, or even greater than, those in single application at lower N rates as basal fertilizer. Overall, these findings suggest that optimizing crop management for N application to enhance yield and NUEs in ratoon-sorghum system is primarily linked to the timing and frequency of N applications rather than simply adjusting N rates. These results are consistent with previous study on rice[72].

Conclusions

The dry matter weight and LAI of MC were significantly higher than those of RC. A higher N rate and splitting application could effectively improve the growth status for both main and ratoon sorghum, as evidenced by the increase in dry matter weight, CGR, LAI, and PP. With an increase in N rate, the main, ratoon, and annual sorghum grain yields increased, but the ratoon and annual yields under 225 kg N ha^{-1} did not show significant improvement compared to those under 150 kg N ha^{-1}. Splitting N application notably enhanced the main, ratoon, and annual yields, especially improved the ratoon yield. At N rate of 150 kg ha^{-1}, increasing the ratio of heading-N fertilizer could significantly increase the ratoon and annual yields. Sorghum cultivar varied in NUEs, with JYN3 demonstrating higher NUEs than GJH1. Decreasing the total N rate from 225 to 75 kg ha^{-1} and increasing the ratio of topdressing to basal N could effectively enhance RE_N, AE_N, and PFP_N in main season and whole year. Our results suggest that T8 (N rate of 150 kg ha^{-1} applied at a ratio of 2: 4: 4) could achieve high yield and NUEs in ratoon-sorghum system in Southwest China.

Methods

Site description

The field experiments were conducted at the Yuxi Crop Experimental Station of Chongqing Academy of Agricultural Sciences, Yongchuan District (105.71°E long., 29.75°N lat., 298 m altitude a.s.l.) (Fig. 2), Chongqing, China in 2019, 2020, and 2021. The soil was in a clay loam texture and the physicochemical properties of the 0–20 cm soil layer are shown in Table 8. Meteorological data were collected from the Yongchuan Meteorological Station (http://yc121.xibuip.cn:8188/) and daily average mean temperature and precipitation were shown in Fig. 3.

Experiment design and treatments

The hybrid sorghum cultivar Jinyunuo3 (JYN3) and the inbred sorghum cultivar Guojiaohong1 (GJH1) were selected for this study. These two sorghum cultivars are extensively cultivated in Southwest China due to their high yield and excellent quality for liquor production. The seeds of JYN3 and GJH1 were obtained from Chongqing Academy of Agricultural Sciences and Sichuan Academy of Agricultural Sciences, respectively. The experiments were arranged in a split-plot design with nitrogen managements as the main plot and sorghum cultivars as subplot, replicated three times. Urea containing 46.4% N was applied at basal, jointing, and heading stages only to the MC, while no fertilizer was applied to the RC. Twelve N treatments ranging from 225 to 75 kg N ha^{-1} were implemented, including a 0 kg N ha^{-1} as control (T0) (Table 9). The ratios of basal, jointing, and heading fertilizer were 1: 0: 0, 5: 5: 0, and 3: 4: 3 for each N rate. Additionally, two more ratios of 2: 5: 3 and 2: 4: 4 were applied at the dose of 150 kg N ha^{-1}. Recommended amounts of phosphorus (75 kg P_2O_5 ha^{-1} in form of superphosphate)

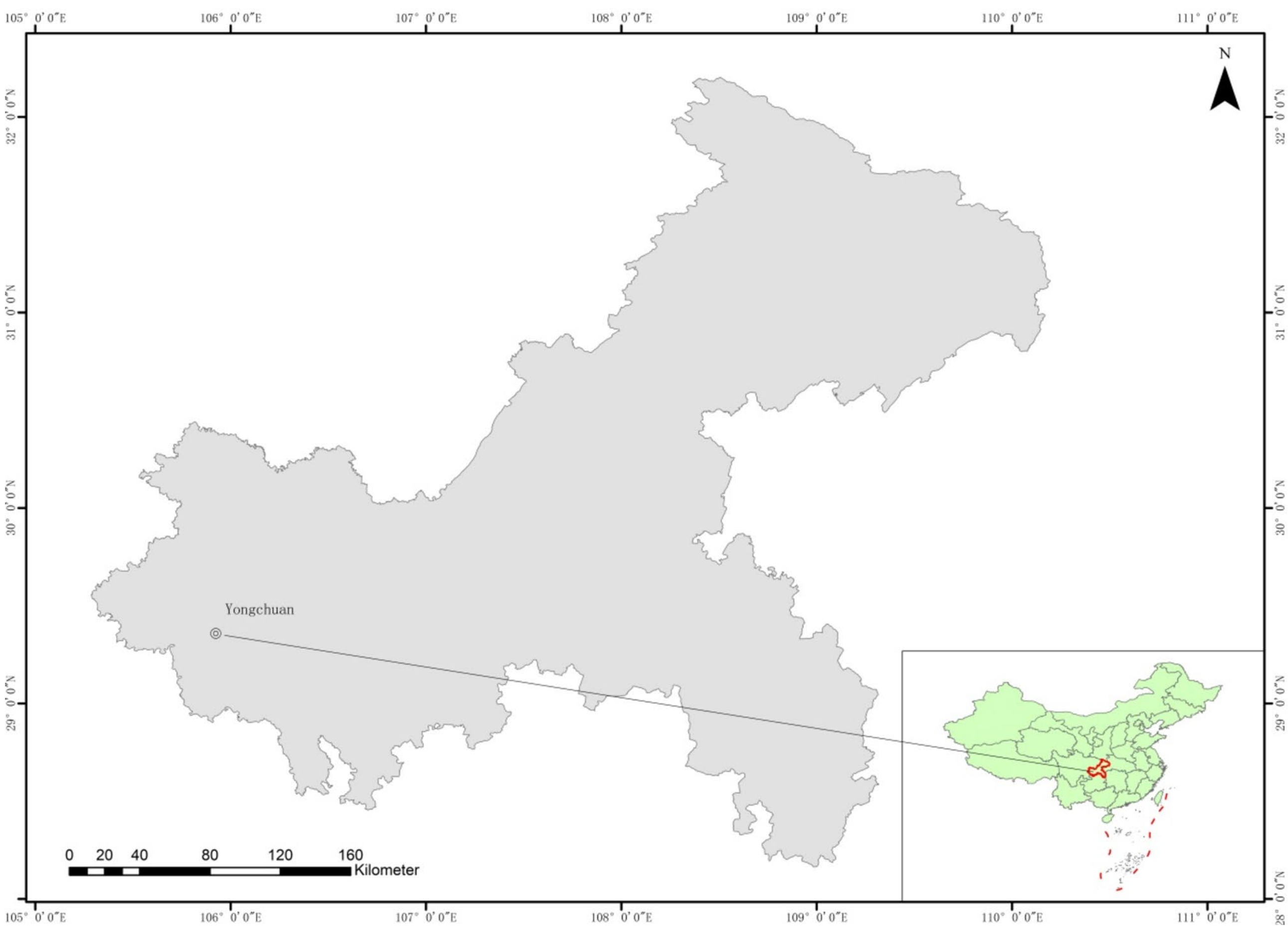

Fig. 2. Location of the study area. The map and inset map in this image were drawn using ArcGIS software (version 10.2, http://www.esri.com/).

Year	2019	2020	2021
pH	4.46	4.67	4.86
Organic matter (g kg^{-1})	23.89	12.32	21.98
Total N (g kg^{-1})	0.76	0.54	0.79
Available (N mg kg^{-1})	98.3	70.78	109.21
Available P (mg kg^{-1})	13.41	8.57	14.08
Available K (mg kg^{-1})	102.69	92.77	67.79

Table 8. The soil physicochemical properties (0–20 cm soil layer) in 2019, 2020, and 2021.

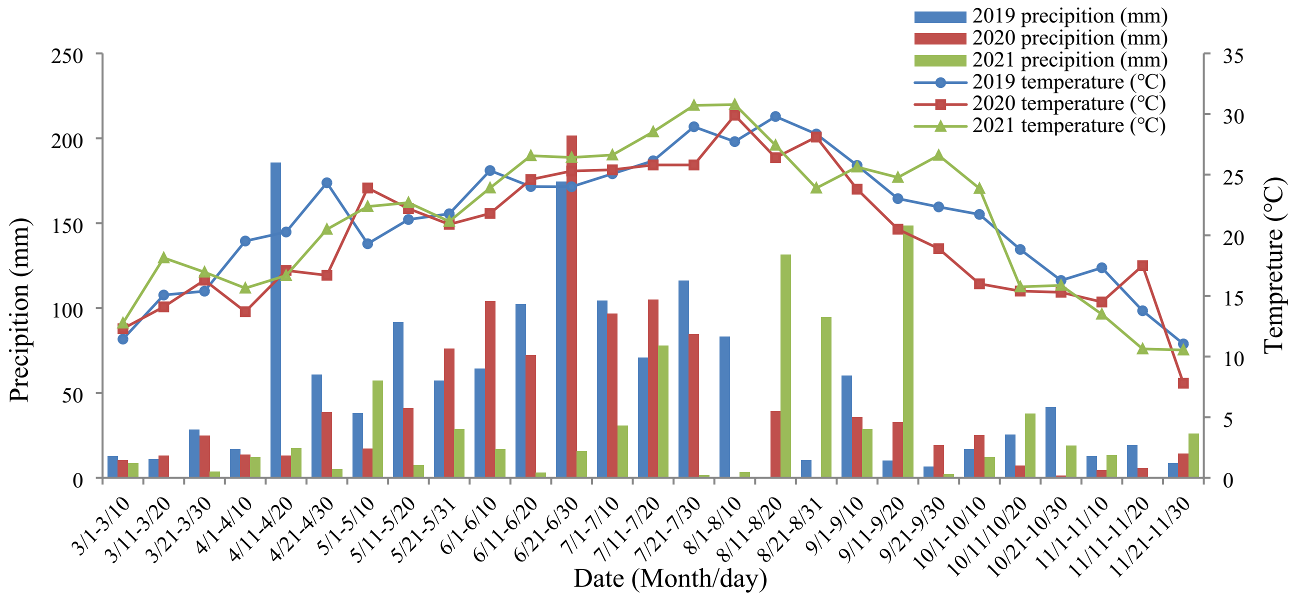

Fig. 3. Precipitation and mean temperature during the growth period (2019–2021) of the ratoon sorghum system at the experimental site.

Treatment	Total N rate	Ratio	Basal	Jointing	Heading
T0	0				
T1	225	1:0:0	225.0	0.0	0.0
T2	225	5:5:0	112.5	112.5	0.0
T3	225	3:4:3	67.5	90.0	67.5
T4	150	1:0:0	150.0	0.0	0.0
T5	150	5:5:0	75.0	75.0	0.0
T6	150	3:4:3	45.0	60.0	45.0
T7	150	2:5:3	30.0	75.0	40.0
T8	150	2:4:4	30.0	60.0	60.0
T9	75	1:0:0	75.0	0.0	0.0
T10	75	5:5:0	37.5	37.5	0.0
T11	75	3:4:3	22.5	30.0	22.5

Table 9. The amount (kg ha^{-1}) of fertilizer N and application ratio for different N treatment.

and potassium (112.5 kg K_2O ha^{-1} in form of potassium sulfate) were applied as basal dressing. Sorghum seeds were sown on 21st March, 4th April, and 8th April in 2019, 2020, and 2021, respectively. The hill seeding was employed with a distance between hills at 38.1 cm, resulting in a planting density of 105 000 plants ha^{-1}. The subplot size was 6.0 m × 5.0 m, containing 12 lines spaced 0.5 m apart.

The seedlings were thinned at the five-leaf stage, with two plants remaining at each hill. The MC was harvested by reaping ears manually on 26th July, 6th August, and 9th August in 2019, 2020, and 2021, followed by cutting the stem at a height of 5–10 cm above the ground. New buds on stubble were removed by hand 10 days after cutting MC, leaving one to grow to maturity in each plant. The ratoon sorghum was harvested on 18th November, 19th November, and 30th November in 2019, 2020, and 2021, respectively.

Plant sampling and measurement

Growth parameters

Field samples were collected at jointing, heading and maturity stages in MC, and at heading and maturity stages in RC. Five representative plants were harvested from each plot. The leaf area of each green leaf was estimated using the coefficient method[73], where leaf area (LA) (cm^2) = leaf length (cm) × maximum width of the leaf (cm) × 0.75. Leaf area index (LAI) was calculated as the total green leaf area divided by the harvest area of each plot[74]. Each plant was partitioned into ear (if present), stem (including sheath), and leaves, and then dried for at least 4 days at 80 ℃ before measuring dry mass. Crop growth rate (CGR) and photosynthetic potential (PP) were calculated using the following formulas[75]:

$$\mathrm{CGR}\left(\mathrm{g\,m^{-2}d^{-1}}\right) = (\mathrm{W_2} - \mathrm{W_1})/(\mathrm{T_2} - \mathrm{T_1}) \tag{1}$$

$$\mathrm{PP}\left(\mathrm{m^2d^{-1}m^{-2}}\right) = 1/2(\mathrm{LAI_1} + \mathrm{LAI_2}) \times (\mathrm{T_2} - \mathrm{T_1}) \tag{2}$$

where, W_1 and W_2 represent the dry weight of samples collected at time T_1 and T_2 respectively; LAI_1 and LAI_2 denote LAI at times T_1 and T_2 correspondingly.

Main, ratoon, and annual yields.

Yield samples were collected at maturity by harvesting all plants in an area of 6 m^2 (4 rows of 3 m length) located in the center of each plot. The crop was harvested and threshed manually. Grain moisture content was promptly determined after threshing using a moisture detector (GAC2100AGRI, Dickey-john, Minneapolis, MN, USA), and the yields were reported at moisture content of 130 g H_2O kg^{-1} on a fresh weight basis.

NUEs

After weighting the dry mass, the samples were milled and sieved to determine the N content using the Kjeldahl method. The NUEs of the apparent recovery rate of applied nitrogen (RE_N), agronomic efficiency of applied nitrogen (AE_N), and partial factor productivity from applied nitrogen (PFP_N) were calculated according to the following formulas[76]:

$$\mathrm{RE_N}(\%) = (\mathrm{U_N} - \mathrm{U_0})/\mathrm{F_N} \tag{3}$$

$$\mathrm{AE_N}\left(\mathrm{kg\,kg^{-1}}\right) = (\mathrm{Y_N} - \mathrm{Y_0})/\mathrm{F_N} \tag{4}$$

$$\mathrm{PFP_N}\left(\mathrm{kg\,kg^{-1}}\right) = \mathrm{Y_N}/\mathrm{F_N} \tag{5}$$

where, Y_0 and U_0 represent the grain yield (kg ha^{-1}) and N accumulation in plants at maturity (kg ha^{-1}) for T0, respectively; Y_N and U_N denote the grain yield (kg ha^{-1}) and N accumulation in plants at maturity (kg ha^{-1}) for N application treatments, respectively; and F_N indicates the N rate (kg ha^{-1}) in N application treatments.

Data analysis

Data were analyzed using analysis of variance (ANOVA), and the means were compared based on the Duncan's New Multiple-Range Test at the 0.05 probability level using Statistical Product and Service Solutions (SPSS, version 19.0)[77]. Tables and figures were created using Microsoft Excel 2007 software for Windows.

Ethics approval

Experimental research and field studies on plants, including the collection of plant material, comply with relevant institutional, national, and international guidelines and legislation.

Data availability

All data generated or analysed during this study are included in this published article.

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Author contributions
Y. Z. and J. H. conceived the experiment and wrote the manuscript, Z.B. L. conducted the experiment, Q.Y. W. and Y.H. L. collected data and analysed the results, Y.Q. Z. and X.C. Z. edited the manuscript, Y. W. analysed the results. All authors reviewed the manuscript.

Competing interests
The authors declare no competing interests.

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